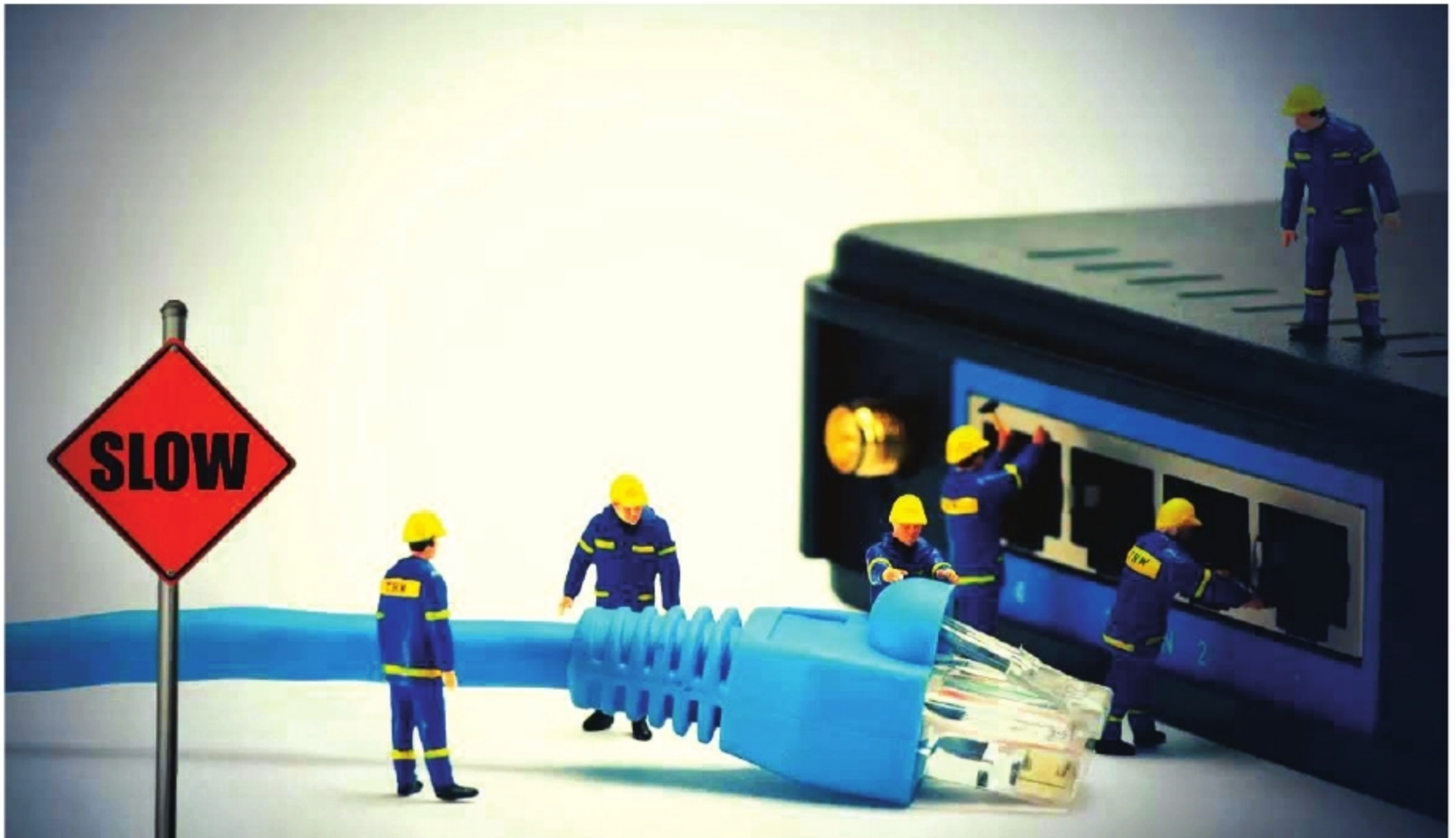


Securing Network Ports with Firewalld

Firewalld provides dynamic monitoring and management of firewalls in RHEL 7. It supports firewall networks or zones that define the trust level of network settings and interfaces. Highly manageable and configurable, it also provides support for IPv4 and IPv6 firewall settings.



Firewalld is the default method for managing host level firewalls in Red Hat Enterprise Linux 7 (RHEL 7). It started off from the `firewall.service` system service, and now manages the Linux kernel netfilter sub-system using low-level iptables, ip6tables, and ebtables commands.

Firewalld separates all incoming traffic into zones, with each zone having its own set of rules. To check which zone to use for an incoming connection, firewalld uses its own logic, using which the first rule that matches wins:

- If the source address of an incoming packet matches a source rule setup for a zone, that packet will be routed through that zone.
- If the incoming interface for a packet matches a filter setup for a zone, that zone will be used.
- Otherwise, the default zone will be used. The default zone is not a separate zone; instead, it points to one of the other zones defined on the system.

Unless overridden by an administrator or a NetworkManager configuration, the default zone for any new network interface will be set to the public zone.

A number of predefined zones are shipped with firewalld, each with its own specified use.

Default configurations for firewalld zones

- **Trusted:** Allows all incoming traffic.
- **Home:** Rejects incoming traffic unless related to outgoing traffic or matching the ssh, mdns, ipp-client, samba-client or dhcpv6-client predefined services.
- **Internal:** Rejects incoming traffic unless related to outgoing traffic or matching the ssh or dhcpv6-client predefined services; this is the default zone for newly added network interfaces.
- **External:** Rejects incoming traffic unless related to outgoing traffic or matching the ssh predefined service. Outgoing IPv4 traffic forwarded through this zone is masqueraded to look like it or that it originated from the IPv4 address of the network outgoing interface.
- **Dmz:** Rejects incoming traffic unless it is related to outgoing traffic or matches the ssh predefined service.
- **Block:** Rejects all incoming traffic unless related to outgoing traffic.